SMARTBRIDGE
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SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	15 July 2026
Team ID	SWTID-2026-2378
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to develop an intelligent **Credit Card Approval Prediction System** using Machine Learning that predicts whether an applicant is eligible for a credit card based on personal, financial, and employment details, enabling faster and more consistent decisionmaking. Key features include a machine learning-based credit model and real-time decisionmaking.

Project Overview	
Objective	The primary objective is to develop an intelligent **Credit Card Approval Prediction System** using Machine Learning that predicts whether an applicant is eligible for a credit card based on personal, financial, and employment details, enabling faster and more consistent decision-making.
Scope	The project focuses on automating the credit card approval process by analyzing applicant information, training a Random Forest classification model, and deploying it as a user-friendly Flask web application for realtime prediction.
Problem Statement	
Description	Traditional credit card approval processes are often time-consuming, require extensive manual verification, and may lead to inconsistent decisions. An automated prediction system is needed to improve efficiency and support data-driven approval decisions.
Impact	Implementing the proposed system will reduce processing time, improve consistency in approval decisions, assist financial institutions in applicant screening, and enhance the overall customer experience through faster responses.

Proposed Solution	
Approach	Develop a Machine Learning-based prediction model using the **Random Forest algorithm**. The model analyzes applicant details such as income, employment, education, family status, credit history, and other relevant features to predict whether the credit card application is likely to be approved or rejected. The trained model is integrated into a Flask web application for real-time predictions.
Key Features	• Machine Learning-based credit card approval prediction using Random Forest. • Web application developed using Flask for real-time prediction. • User-friendly interface for entering applicant details. • Fast and consistent prediction results. • Model trained using historical applicant and credit history data.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm

Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv